

Auteur: Michael Livchits
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$$\begin{aligned} & \int \coth^2 x \cosh x dx \\ &= \int \frac{\cosh^2 x}{\sinh^2 x} \cosh x dx \\ &= \int \frac{1 + \sinh^2 x}{\sinh^2 x} \cosh x dx \\ &= \int \frac{\cosh x}{\sinh^2 x} dx + \int \cosh x dx \\ &= \int \frac{\cosh x}{\sinh^2 x} dx + \sinh x + C \end{aligned}$$

$$u = \sinh x$$

$$du = \cosh x dx$$

$$\begin{aligned} &= \int \frac{\cosh x}{u^2 \cosh x} du + \sinh x + C \\ &= \int \frac{du}{u^2} + \sinh x + C \\ &= -\frac{1}{u} + \sinh x + C \\ &= -\frac{1}{\sinh x} + \sinh x + C \end{aligned}$$

References